

Leo Strangman

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leostrangman.com | linkedin.com/in/leostrangman

Education:

Worcester Polytechnic Institute | Worcester, MA |

2022-2026

- Bachelor of Science | Major: Computer Science | Minor: Data Science | GPA: 3.88 | Graduated May 2026
- Relevant coursework: Software Engineering, Machine Learning, Artificial Intelligence, Operating Systems, Machine Organization, Human-Computer Interaction, OOP, Algorithms, Webware, and Mobile and Ubiquitous Computing

Skills:

Languages: Python, JavaScript, TypeScript, C, C++, Java, HTML/CSS, SQL, Kotlin, Bash

Frameworks/Libraries: React, Node.js, Flask, Astro, Svelte, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib, Tailwind, scikit-learn

Tools/Platforms: Git/GitHub, Docker, AWS, Azure, PostgreSQL, SQLite, Ubuntu/Linux, Figma, Blender, Anaconda, Vercel, MongoDB, VirtualBox, GDB, VSCode, JetBrains IDEs, Android Studio

Development Practices: Agile/Scrum, CI/CD, Version Control, Object-Oriented Programming, Test-Driven Development, RESTful APIs, Full-Stack Development, Software Architecture, Database Design, API Development, Unit Testing, Debugging, Troubleshooting

Work Experience:

Software Engineering Intern | Massachusetts General Hospital | Boston, MA |

2022-2025

- Front and back-end website development for two specialized research departments, the Neural Systems Group and the Center for Space Medicine Research using Astro, Tailwind CSS, Postgres, and Vercel
- Automated analysis of data collected with biomedical research equipment using Python to increase analysis efficiency by over 80% by reducing manual input
- Prototyped housing and sensor components for a novel biomedical monitoring and brain imaging system to be used for neurophysiology research aboard the International Space Station
- Contributed original CAD designs based on initial structures, which were adopted by the team for future iterations

Projects:

Tablet Interface for Robot Assisted Laser Surgery | C, Python, TypeScript, HTML, Figma |

2025-2026

- Developed a novel tablet-stylus interface for a 7-DOF surgical robot, allowing surgeons to precisely resolve 0.5mm sutures on 1mm vessels in real time via a synchronized WebRTC live video feed, validated by a user study.
- Designed a robust hardware-software bridge utilizing WebSockets to translate custom user-drawn toolpaths into executable trajectories with sub-millimeter accuracy and toolpath simulation for planned execution.
- Integrated thermal telemetry and visual overlays, displaying live robot positions and dynamic heat maps directly on the frontend feed of the surgical site with an HTML5 Canvas rendering system for advanced spatial awareness.
- Programmed critical safety protocols, including user-defined exclusion zones and automatic path-validation, improving surgical decision-making and decreasing procedural complexity and cognitive load.

School Benchmarking Platform – Osprey Software | React, Node.js, Express.js, MongoDB |

2026

- Architected and developed a full-stack MERN benchmarking web application for tracking institutional KPIs
- Engineered optimized REST APIs to execute database operations, reducing server response time for complex data
- Integrated Auth0 authentication to establish secure user access and role-based route protection
- Designed high-density data entry forms with real-time validation and asynchronous draft-saving to prevent data loss

leostrangman.com | Astro, Tailwind, TypeScript, JavaScript, Vercel |

2024-2025

- Designed and developed an Astro-based responsive and engaging personal portfolio website with complex animations

ResearchApp | Python, Flask, HTML, JavaScript, SQLAlchemy |

2024

- Created an application used to post and apply to research positions within a university community of 6,000+ students, integrating RESTful APIs to manage user data, research listings, and applications
- Utilized the Agile software development methodology to collaborate effectively within an 8-person team
- Designed the database schema with SQLAlchemy ORM and integrated the Flask backend with the frontend using Jinja2 to dynamically update pages with a templating system

Threads, Memory Management, and Scheduling | Ubuntu, VirtualBox, C |

2024

- Engineered a multi-threaded simulation of a virtual sports field in C, implementing decentralized thread coordination to manage concurrent team and player interactions without a centralized scheduler
- Designed and implemented a custom paging system with memory-mapped addressing, supporting map, store, and load operations for simulated virtual memory management

Exploiting Programs in C and Assembly | Assembly (Linux x86), C, GDB, pwndbg |

2023

- Decompiled C code to exploit vulnerabilities via reverse engineering, crafted inputs, and control flow manipulation
- Executed stack-based buffer overflow attacks and implemented ROP chains to bypass security mechanisms